LLDA Quarterly Water Quality Monitoring Report
Laguna Lake and Tributary Rivers
April to June, 2016



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Laguna Lake Stations

Monitoring Stations	Water Quality Parameters											
	pH (units)			Ammonia (mg/L)			Nitrate (mg/L)			Inorganic Phosphate (mg/L)		
	April	May	June	April	May	June	April	May	June	April	May	June
Stn. I (Central West Bay)	8.7	8.3	9.4	0.152	0.236	0.063	0.065	0.001	0.065	0.125	0.147	0.108
Stn. II (East Bay)	8.5	8.7	9.3	0.085	0.081	0.092	0.046	0.009	0.050	0.071	0.037	0.181
Stn. IV (Central Bay)	8.7	8.3	9.6	0.129	0.050	0.051	0.039	0.002	0.051	0.104	0.103	0.149
Stn. V (Northern West Bay)	8.9	8.2	9.0	0.083	0.349	0.336	0.087	0.059	0.069	0.102	0.164	0.190
Stn. VIII (South Bay)	8.5	8.3	9.4	0.094	0.055	0.054	0.081	0.001	0.684	0.084	0.077	0.095
Stn.XV (San Pedro)	8.3	8.2	8.9	0.088	0.074	0.067	0.128	0.019	0.770	0.103	0.069	0.120
Stn.XVI (Sta. Rosa)	8.3	8.3	9.0	0.081	0.054	0.042	0.102	0.036	0.165	0.087	0.067	0.087
Stn.XVII (Sanctuary)	8.6	8.1	9.3	0.058	0.079	0.047	0.088	0.051	0.372	0.089	0.066	0.073
Stn.XVIII (Pagsanjan)	8.7	9.4	9.4	0.068	0.056	0.031	0.096	0.004	0.798	0.031	0.004	0.120

Tributary River Stations

Station	Monitoring Stations	Water Quality Parameters								
		pH (units)			Nitrate (mg/L)			Inorganic Phosphate (mg/L)		
		April	May	June	April	May	June	April	May	June
1	Marikina River	7.1	7.2	7.3	0.001	0.081	0.001	1.106	0.566	0.512
2	Bagumbayan River (Taguig)	6.4	7.5	7.4	0.062	0.001	0.028	2.310	3.140	2.160
3	Buli Creek (Taguig)	7.1	6.6	7.2	0.025	0.056	0.076	2.450	2.200	1.720
4	Mangangate River (Muntinlupa) - Downstream	-	7.5	7.7	-	0.001	0.055	-	1.270	1.120
4U	Mangangate River (Muntinlupa) - Upstream	7.8	7.8	7.6	0.149	0.291	0.141	1.610	1.190	0.620
5	Tunasan River (Muntinlupa) - Downstream	7.5	8.0	6.5	0.118	0.113	0.088	1.700	4.760	0.249
5U	Tunasan River (Muntinlupa) - Upstream	7.9	7.1	8.0	0.093	0.405	0.655	1.610	1.750	1.000
6	San Pedro River (T_2)	7.9	7.9	7.6	0.033	0.777	0.916	1.952	1.360	1.026
7	Biñan River	7.9	7.9	7.5	0.071	0.000	0.070	1.408	1.150	0.692
8	Sta. Rosa River - Downstream	7.7	7.8	7.5	0.063	0.518	0.001	0.662	0.142	0.421
8M	Sta. Rosa River - Midstream	7.3	7.8	7.5	0.122	0.510	0.022	0.710	0.189	0.607
8U	Sta. Rosa River - Upstream	8.0	8.1	7.9	0.921	1.030	1.024	0.624	0.582	0.502
9	Cabuyao River	7.7	7.7	7.4	0.034	0.722	0.001	1.330	0.437	0.838
10	San Cristobal River (T_3)	6.8	7.0	6.8	0.001	0.042	0.029	0.878	0.345	0.587
11	San Juan River (T_5)	7.7	7.6	7.5	3.940	1.330	3.280	0.963	0.818	1.049
12	Molawin Creek (Los Baños)	7.7	7.7	7.5	0.546	0.062	0.556	0.458	0.634	0.423
13	Bay River (T_9)	8.1	8.1	8.0	0.607	0.732	0.547	0.308	0.304	0.296
14	Pila River	7.4	7.4	7.5	0.149	0.266	0.141	0.102	0.143	0.204
15	Sta. Cruz River (T_6)	7.7	7.7	7.5	0.437	0.762	0.504	0.139	0.187	0.215
16	Pagsanjan River (T_8)	7.7	7.7	7.6	0.284	1.000	0.217	0.092	0.133	0.189
17	Pangil River - Downstream	7.8	7.8	7.3	0.077	0.091	0.231	0.102	0.133	0.075
17U	Pangil River - Upstream	7.3	7.6	7.2	0.155	0.030	0.156	0.069	0.124	0.039
18	Siniloan River	7.9	7.5	6.9	0.142	0.069	0.355	0.274	0.408	0.101
19	Sta. Maria River - Downstream	8.3	7.6	7.5	0.142	0.118	0.281	0.124	0.170	0.067
19U	Sta. Maria River - Upstream	8.3	8.7	8.2	0.019	0.004	0.152	0.067	0.106	0.100
20	Jala-jala River	-	-	-	-	-	-	-	-	-
21	Pililla River	7.2	7.3	7.2	0.039	0.013	0.076	0.280	0.443	0.280
22 A	Tanay River - Brgy. Wawa	7.8	8.1	8.1	0.001	0.105	0.001	0.111	0.120	0.073
22 B	Tanay River - Daranak Falls	8.2	8.1	8.4	0.124	0.137	0.001	0.001	0.018	0.031
22 C	Tanay River- Cuyambay	7.8	7.8	7.9	0.001	0.780	0.001	0.001	0.022	0.001
23	Baras River	7.7	7.9	7.6	0.001	0.001	0.080	0.105	0.074	0.140
24	Morong River - Downstream	-	-	7.9	-	-	0.830	-	-	0.984
24U	Morong River - Upstream	8.1	7.9	8.0	0.670	0.185	0.502	1.492	2.200	0.996
25	Manggahan Floodway (Taytay)	7.5	7.3	7.3	0.032	0.001	0.066	1.586	0.757	1.102
26	Sapang Baho River (Cainta)	7.3	7.3	7.4	0.001	0.001	0.001	1.572	0.844	0.718
27	Angono River	7.4	7.5	7.5	0.072	0.299	0.885	0.506	0.401	0.347

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Laguna Lake Stations

Monitoring Stations	Water Quality Parameters*								
	BOD (mg/l)			Dissolved Oxygen (mg/l)			Total coliforms (MPN/100ml)**		
	April	May	June	April	May	June	April	May	June
Stn. I (Central West Bay)	1	1	8	8.6	6.4	14.4	144	144	85
Stn. II (East Bay)	1	3	2	8.3	8.4	8.3	627	945	973
Stn. IV (Central Bay)	1	1	3	8.6	6.8	9.5	125	117	117
Stn. V (Northern West Bay)	1	2	3	7.5	7.1	9.5	125	102	363
Stn. VIII (South Bay)	1	1	3	8.3	7.9	8.9	428	620	709
Stn.XV (San Pedro)	2	3	3	8.6	6.9	7.6	201	185	208
Stn.XVI (Sta. Rosa)	*1*	*2*	*2*	*8.1*	*7.7*	*7.7*	*165*	*107*	83
Stn.XVII (Sanctuary)	1	2	3	8.3	7.5	8.9	294	86	35
Stn.XVIII (Pagsanjan)	1	4	4	8.5	10.0	9.1	110	51	26

Notes: * DENR Class C Water Quality Criteria:
Biochemical Oxygen Demand (BOD) - 10 mg/l
Dissolved Oxygen (DO) - 5 mg/l
Total Coliforms - 5000 MPN/100 ml
** Based on computed geomean

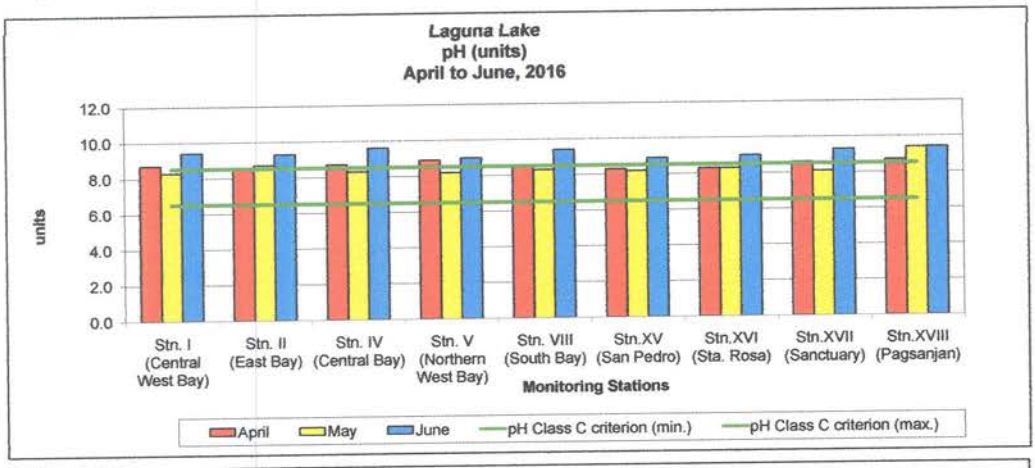
Tributaty Rivers Stations

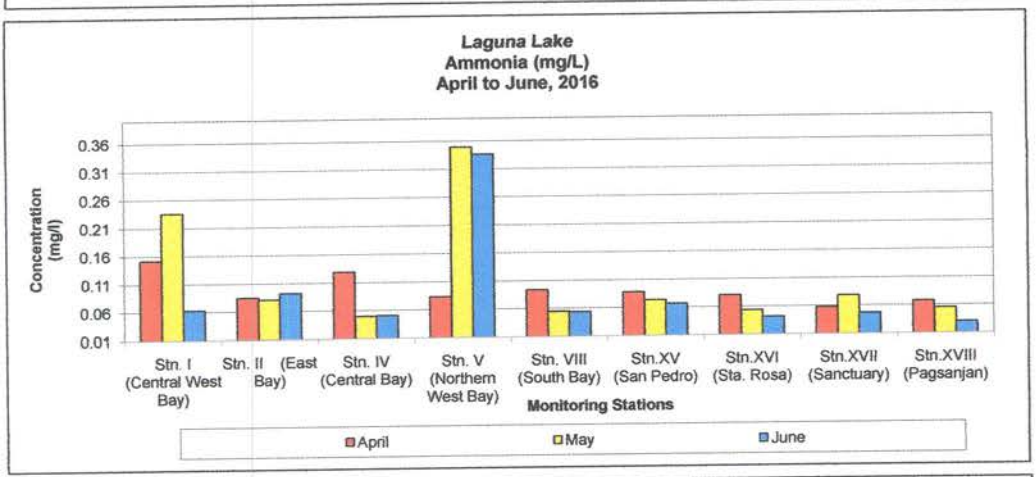
Station	Monitoring Stations	Water Quality Parameters*								
		BOD (mg/l)			Dissolved Oxygen (mg/L)			Total coliforms (MPN/100ml)**		
		April	May	June	April	May	June	April	May	June
1	Marikina River	37	43	14	0.05	0.05	0.20	3,500,000	3,500,000	3,500,000
2	Bagumbayan River (Taguig)	650	264	245	0.05	0.05	0.05	23,835,918	23,835,918	13,304,828
3	Buli Creek (Taguig)	211	982	300	0.05	0.05	0.05	11,063,653	13,304,828	13,304,828
4	Mangangate River (Muntinlupa) - Downstream	-	96	103	-	0.05	0.05	13,000,000	34,583,233	17,424,121
4U	Mangangate River (Muntinlupa) - Upstream	12	10	7	3.60	4.80	3.60	1,089,599	955,079	562,386
5	Tunasan River (Muntinlupa) - Downstream	390	487	555	0.05	0.05	0.50	25,635,952	59,189,090	70,692,447
5U	Tunasan River (Muntinlupa) - Upstream	7	9	8	4.00	3.10	5.10	462,092	259,412	391,402
6	San Pedro River (T_2)	12	9	7	0.05	-	0.05	380,809	446,530	391,402
7	Biñan River	37	25	25	0.05	0.20	0.90	5,581,486	5,581,486	6,712,134
8	Sta. Rosa River - Downstream	26	16	27	0.05	-	0.05	3,284,099	2,098,411	5,122,317
8M	Sta. Rosa River - Midstream	64	10	15	0.05	-	2.80	6,712,134	11,139,813	11,139,813
8U	Sta. Rosa River - Upstream	7	8	6	6.30	6.10	6.40	1,330,483	1,106,365	488,103
9	Cabuyao River	9	37	33	4.50	0.05	0.05	6,929,913	6,929,913	6,929,913
10	San Cristobal River (T_3)	132	93	100	0.05	0.30	0.05	48,303,059	48,303,059	30,863,780
11	San Juan River (T_5)	4	13	8	6.00	4.60	4.80	590,307	765,628	685,639
12	Molawin Creek (Los Baños)	3	3	3	3.50	2.10	2.70	44,145	26,454	37,366
13	Bay River (T_9)	3	2	4	6.10	5.90	5.60	493,301	441,602	639,576
14	Pila River	3	3	3	3.20	2.70	1.90	71,464	84,371	46,209
15	Sta. Cruz River (T_6)	1	1	1	6.00	5.80	4.20	42,951	50,363	110,653
16	Pagsanjan River (T_8)	1	2	1	6.70	5.90	6.10	170,173	135,066	81,382
17	Pangil River - Downstream	2	3	1	8.80	8.10	6.20	167,419	274,970	134,359
17U	Pangil River - Upstream	1	3	1	7.50	7.40	7.70	30,550	59,189	23,314
18	Siniloan River	4	6	3	2.50	3.30	4.00	97,576	147,717	125,975
19	*Sta. Maria River - Downstream*	*1*	*4*	*3*	*6.80*	*5.70*	*4.30*	*97,576*	*187,918*	*140,475*
19U	Sta. Maria River - Upstream	1	4	1	6.90	8.60	7.70	15,662	41,626	36,487
20	Jala-jala River	-	-	-	-	-	-	-	-	-
21	Pililla River	7	13	8	1.70	2.20	1.20	211,407	401,213	770,294
22 A	Tanay River - Brgy. Wawa	6	28	8	10.10	9.60	9.30	41,555	15,723	10,615
22 B	Tanay River - Daranak Falls	1	1	1	8.70	7.40	5.00		14,930	11,800
22 C	Tanay River- Cuyambay	2	1	1	7.50	7.00	6.80	3,088	3,088	13,305
23	Baras River	9	7	11	9.10	0.05	8.90	76,242	69,721	177,006
24	Morong River - Downstream	-	-	13	-	-	5.20	65,498	130,000	240,000
24U	Morong River - Upstream	17	12	14	11.00	2.80	5.00	27,386	17,340	20,472
25	Manggahan Floodway (Taytay)	18	22	16	3.80	0.05	0.05	420,555	333,795	646,713
26	*Sapang Baho River (Cainta)*	*26*	*19*	*22*	*0.05*	*0.05*	*0.05*	*6,712,134*	*5,808,786*	*8,016,632*
27	Angono River	10	6	9	1.50	2.50	3.50	3,682,066	1,594,775	1,300,000

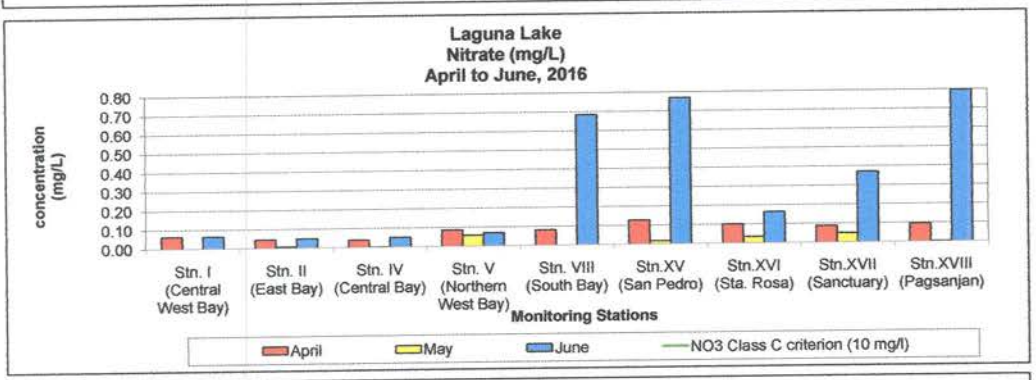
Notes: * DENR Class C Water Quality Criteria:
Biochemical Oxygen Demand (BOD) - 10 mg/l
Dissolved Oxygen (DO) - 5 mg/l
Total Coliforms - 5000 MPN/100 ml
** Based on computed geomean
*** Stations renamed were Tanay Upstream to Tanay River-Daranak Falls and Tanay Downstream to Tanay River - Brgy. Wawa after Tanay - Cuyambay monitoring station was added starting in February 2016

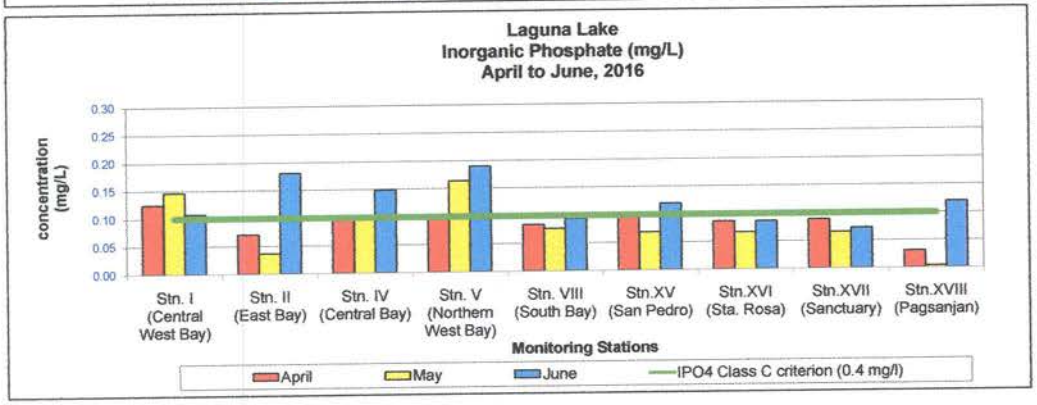
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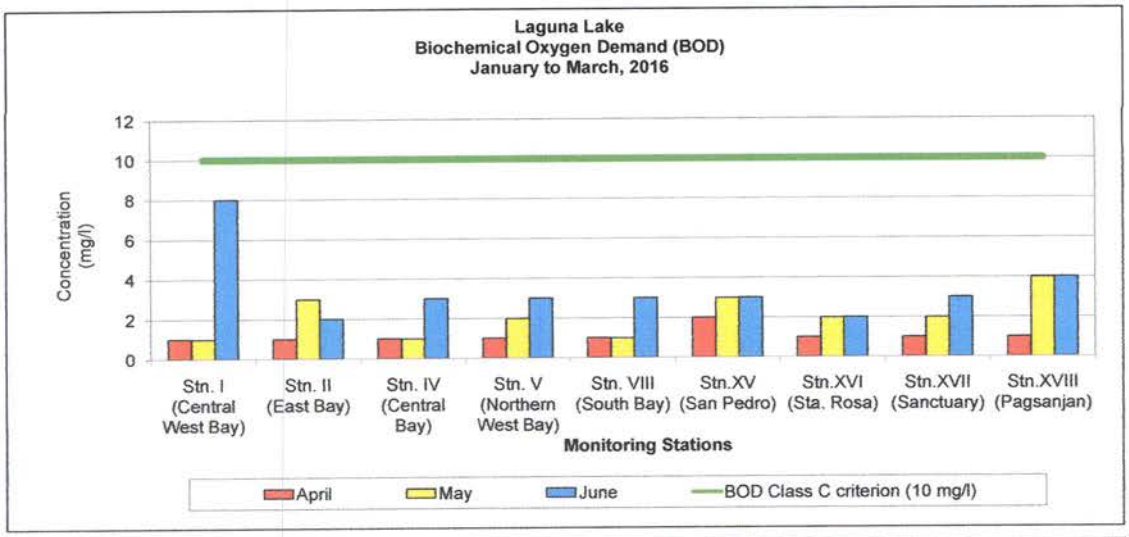
Laguna Lake

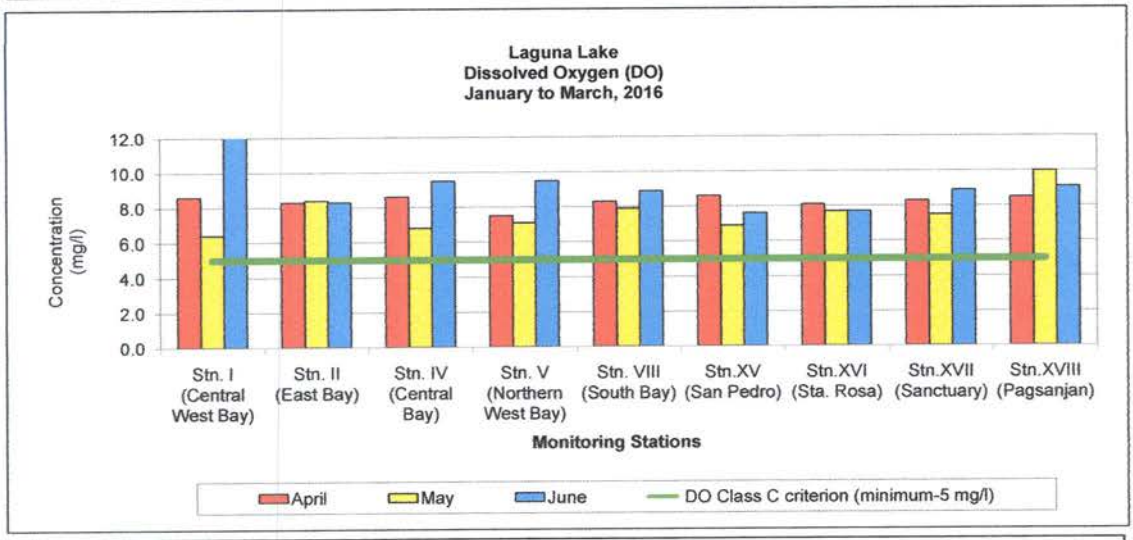


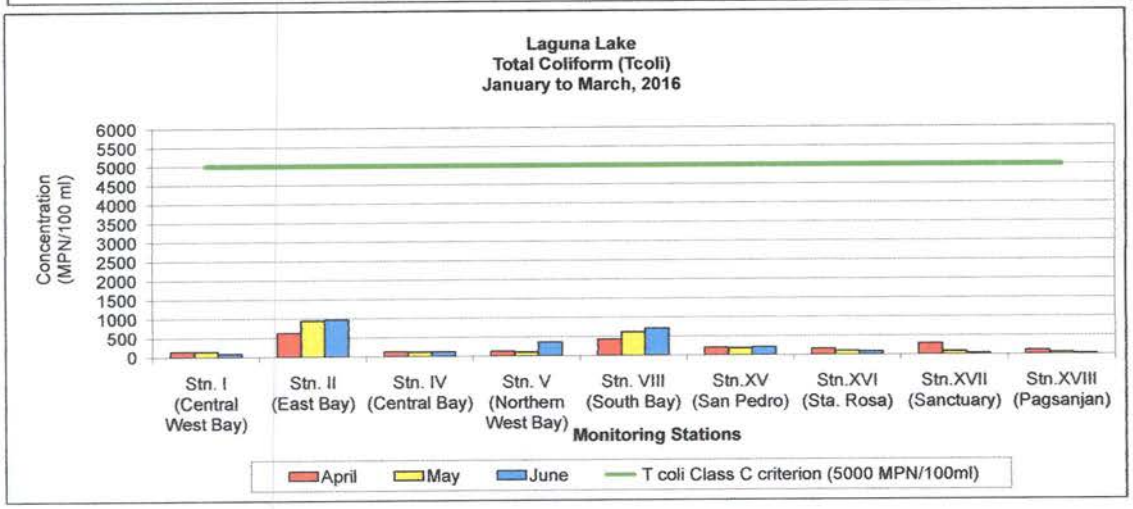







Laguna Lake
Biochemical Oxygen Demand (BOD)
January to March, 2016
Concentration (mg/l)
12
10
8
6
4
2
0
Stn. I (Central West Bay)
Stn. II (East Bay)
Stn. IV (Central Bay)
Stn. V (Northern West Bay)
Stn. VIII (South Bay)
Stn.XV (San Pedro)
Stn.XVI (Sta. Rosa)
Stn.XVII (Sanctuary)
Stn.XVIII (Pagsanjan)
Monitoring Stations
April
May
June
BOD Class C criterion (10 mg/l)


Laguna Lake
Dissolved Oxygen (DO)
January to March, 2016
Concentration (mg/l)
12.0
10.0
8.0
6.0
4.0
2.0
0.0
Stn. I (Central West Bay)
Stn. II (East Bay)
Stn. IV (Central Bay)
Stn. V (Northern West Bay)
Stn. VIII (South Bay)
Stn.XV (San Pedro)
Stn.XVI (Sta. Rosa)
Stn.XVII (Sanctuary)
Stn.XVIII (Pagsanjan)
Monitoring Stations
April
May
June
DO Class C criterion (minimum-5 mg/l)


Laguna Lake
Total Coliform (Tcoli)
January to March, 2016
Concentration (MPN/100 ml)
6000
5500
5000
4500
4000
3500
3000
2500
2000
1500
1000
500
0
Stn. I (Central West Bay)
Stn. II (East Bay)
Stn. IV (Central Bay)
Stn. V (Northern West Bay)
Stn. VIII (South Bay)
Stn.XV (San Pedro)
Stn.XVI (Sta. Rosa)
Stn.XVII (Sanctuary)
Stn.XVIII (Pagsanjan)
Monitoring Stations
April
May
June
T coli Class C criterion (5000 MPN/100ml)

FINDINGS:

A. On Biochemical Oxygen Demand (BOD):

Laguna Lake:

- For the 2nd quarter of 2016, all of the nine (9) lake stations passed the 10 mg/L BOD criterion set for Class C waters by the DENR.
- The BOD concentrations for the 2nd quarter ranged only from 1 to 8 mg/L.

Tributary Rivers:

- During the 2nd quarter , the results of the BOD analysis showed that out of the 36 tributary stations monitored, only 15 stations consistently conformed with the Class C criterion for BOD and these were Stns. 5U, 8U, 12 to 19U, 22B, 22C and 27.
- All of the monthly BOD concentrations in Stns. 1, 2, 3, 4, 5, 7, 8, 10, 24, 24U, 25 and 26 failed the 10 mg/L Class C criterion.
- The BOD concentrations in Stns. 4U and 6 conformed with the criterion in May and June but failed in April while Stns. 8M and 9 passed the criterion only in May and April respectively.
- In April and June, Stns. 11, 21 and 22A conformed with the Class C criterion for BOD but failed in May whereas Stn. 23 conformed with the criterion in April and May but failed in June.
- No sampling was undertaken in Jala-jala River (Stn.20) because the river was dry from April to March.
- The BOD concentrations for the 2nd quarter of 2016 ranged from 1.0 to 982 mg/L with the highest noted in Buli Creek (Stn. 3) in May.

B. On Dissolved Oxygen (D.O.):

Laguna Lake:

- All of the nine (9) lake stations conformed with the DENR Class C criterion for D.O. set at a minimum 5 mg/L.
- The lowest D.O. concentration at 6.4 mg/L was measured in Stn.I in May while the highest concentration was at 10.0 mg/L in Stn. XVIII also in May.

Tributary Rivers:

- From April to June 2016, only 9 stations out of the 36 tributary stations monitored, consistently passed the Class C criterion for DO and these were Stns. 8U, 13, 16, 17, 17U, 19U, 22A, 22B and 22C.
- The monthly concentrations for DO in the nineteen (19) tributary stations always failed the 5.0 mg/L Class C criterion and these were Stns. 1, 2, 3, 4, 4U, 5, 6, 7, 8, 8M, 9, 10, 12, 14, 18, 21, 25, 26 and 27.
- The DO concentrations complied with the criterion in Stns. 5U and 11U in June and April, in Stn. 15 in April, in Stn. 19 in May and in Stns. 23 and 24U in April and June.
- The DO concentration in Morong River-Downstream (Stn. 24) which was taken only in June conformed with the criterion.
- The recorded DO concentrations with tributaries ranged from 0.05 to 11.0 mg/L.

C. On Total Coliform (T. Coli) :

Laguna Lake:

- For the 2nd quarter of 2016, the monthly geomeans for total coliform in all of the nine (9) lake stations conformed with the DENR Class C criterion of 5000 MPN/100ml.
- The monthly geomeans for T.coli in the lake ranged from 26 MPN/100ml to 973 MPN/100ml.

Tributary Rivers:

- The monthly geomean T. coli concentrations in all the tributary stations failed the Class C criterion except in Tanay River – Cuyambay (Stn. 22C) in April and May both with geomean concentration of 3,088 MPN/100ml..
- The highest monthly geomean for total coliform was recorded in San Cristobal River (Stn.10) with geomean of 48,303,059 MPN/100ml in April and May.

D. On pH :

Laguna Lake:

- For the 2nd quarter of 2016, the recorded pH values in the lake ranged from 8.1 to 9.6.
- The DENR Class C criteria for pH is 6.5 to 8.5. The pH levels in the lake exceeded the maximum criterion of 8.5 in all Stations in June, in Stns. I, IV, V, XVII and XVIII in April, and in Stns. II and XVIII in May.
- High pH values could be due to high photosynthetic activity of algae at the time of sampling.

Tributary Rivers:

- The recorded pH in tributary stations during the second quarter of 2016 ranged from 6.4 to 8.7. The lowest pH concentration was noted in Bagumbayan River (Stn.2) in April while the highest pH value was measured in Sta. Maria River – Upstream (Stn.19U) in May.

E. On Ammonia:

Laguna Lake:

- DAO 34 has no Class C criterion for ammonia.
- The ammonia concentrations in the entire lake ranged from 0.031 to 0.349mg/L.

F. On Nitrate:

Laguna Lake:

- For the second quarter of 2016, all the nine (9) lake stations conformed with the DENR Class C criterion set at 10 mg/L.
- The nitrate concentrations in all of the nine (9) lake stations ranged from 0.001 to 0.798 mg/L.

Tributary Rivers:

- For the second quarter, the nitrate concentrations in all 36 tributary stations conformed with the 10 mg/L Class C criterion for nitrate.
- Measured nitrate concentrations in the tributary stations ranged from 0.001mg/L to 3.940 mg/L. The highest nitrate concentration was noted in San Juan River (Stn. 11) in April.

G. On Inorganic Phosphate:

Laguna Lake:

- Out of the nine (9) lake stations, only three (3) lake stations consistently conformed with the 0.1 mg/L DENR Class C criterion from April to June and these were Stns. VIII, XVI and XVII. The phosphate concentration in Stn.I, Stn. IV and Stn. V always failed to meet the Class C criterion.
- The lowest inorganic concentration was at 0.004 mg/L in Stn. XVIII in May while the highest inorganic phosphate was at 0.190 mg/L in Stn. V in June .

Tributary Rivers:

- For the tributary rivers the DENR Class C criterion is set at 0.4 mg/L. Out of 36 monitored stations, only twelve (12) tributary river stations consistently conformed with the criterion and these were Stns. 13 to 17U, 19, 19U, 22A, 22B, 22C and 23
- Sixteen (16) tributary stations always failed the criterion, namely; Stns. 1 to 4U, 5U to 7, 8U, 9, 11, 12, 24, 24U, 25 and 26.
- The Class C criterion was exceeded in Stns. 5 and 27 in April and May, in Stns. 8, 8M and 10 in April and June and in Stns. 18 and 21 in May.
- The levels of inorganic phosphate in tributaries ranged from 0.001 mg/L to 4.760 mg/L.

Comparatively, the water quality of the lake in the 1st quarter was better compared to the 2nd quarter as based on measured pH and Inorganic Phosphate levels.

In the 1st quarter, all of the nine (9) lake stations conformed with the Class C criteria for pH while in 2nd quarter, high pH values were recorded in the lake in all stations especially in June where all stations exceeded the Class C criteria. The inorganic phosphate concentration in the 1st quarter is better than in 2nd quarter because eight (8) stations consistently conformed with 0.1 mg/L Class C criterion for inorganic phosphate with only Stn. V (Northern West Bay) exceeded criterion while in 2nd quarter, only (3) stations namely; Stns. VIII, XVI and XVII consistently conformed with criterion.

The water quality in the tributary was better in the 1st quarter than in the 2nd quarter because more stations conformed with Class C criteria for BOD and dissolved oxygen. In the 1st quarter, seventeen (17) tributaries consistently conformed with criterion for BOD criterion while only fifteen (15) tributaries in 2nd quarter. Out of the 36 tributaries, eleven (11) tributaries consistently passed the Class criterion for DO in the 1st quarter as compared to only nine (9) tributariesin the 2nd quarter.